

Safe operating envelope for limiting actuation of electric trailer in tractor-semitrailer combination

Axel Hansson¹, Erik Andersson², Leo Laine³, Maliheh Sadeghi Kati⁴, Umur Erdinc⁵ and Mats Jonasson⁶

Abstract—This paper shows how a safe operating envelope based on simulated data can be implemented in an allocation-type control system for a two-unit heavy vehicle. The vehicle in focus is a tractor semi-trailer combination for which firstly safe driving conditions are defined and then control laws and systems are designed with respect to set safe driving conditions. The proposed system is evaluated in a simulation environment and is verified to, during some typical use cases, avoid violating the definitions of safe driving while also minimizing energy losses with regard to some actuation prioritization laws. The system is constructed in a modular way which allows for changes in vehicle configurations and changes in specifications of installed actuators.

I. INTRODUCTION

Research of battery electric vehicles (BEVs) is a popular topic in the heavy vehicle industry as it serves as an environmentally sustainable alternative to combustion engines run with fossil fuels. Several companies in the heavy truck industry aim at in the future only to sell and produce trucks which only use electrical power as the main source of propulsion [10]. One of the many issues with battery operated heavy trucks is to maintain an equivalently long range as their fossil fuel powered counterparts. A novel solution to extending the range of the vehicle is to add more batteries to the vehicle, thus increasing the total battery energy capacity on board. If batteries and electrical motors are installed in the trailer of a tractor semi-trailer combination, the trailer could be used as a propulsive complement to the vehicle. This allows for the possibility of using the electrified trailer (e-trailer) on both tractors with internal combustion engines and on BEV-tractors. The e-trailer could also be used to recharge the batteries through regenerative braking, thus preventing wasting energy on mechanical brakes.

Using a trailing unit, such as a semi-trailer, for propulsion may however contribute negatively to the stability of the combination, particularly the yaw stability. There is an increased risk of unwanted and dangerous vehicle behaviors such as jackknifing and trailer swing. In order to prevent such unsafe behaviors a sophisticated control strategy is needed. This can be handled by using an actuator limiting safe operating envelope (SOE) in conjunction with control allocation

algorithms. Control allocation with a weighted least squares (WLS) solver can be used to solve the optimization issue of distributing actuator signals on a heavy multi-unit vehicle which is a popular and well studied method [2] [3] [4] [5] [6].

A set of applied actuator signals together with environmental parameters and vehicle states which, by some logic, can predict or ensure safe driving with a high probability is called a safe operating envelope. The use of SOEs as a limiting function regarding the capabilities of heavy vehicles and ensuring safe driving has been studied and researched using model predictive controllers on yaw rate and wheel sideslip angle [7] [8] [9]. It has however not been done using off-line generated simulated experimental data with different and more versatile envelope parameters to limit an electric trailer on a two-unit heavy vehicle. Such an envelope as presented in this paper can therefore determine what driver inputs will yield safe vehicle motion and which will not with respect to vehicle states and environmental parameters.

The first step of finding the SOE is to define proper definitions of what is considered to be safe vehicle motion. This will be declared in Section II. Then the definition of the SOE will be described in Section III followed by a description of the control system and how the SOE is integrated in Section IV. Results of the implemented system for two typical driving scenarios are presented in Section V and conclusions about the feasibility of the suggested system and possible embodiments of the same are presented in Section VI.

II. DEFINING SAFE VEHICLE MOTION

By investigating and controlling the acceleration and braking requests from the truck driver when driving with an e-trailer, safe driving can be ensured. Doing this will prevent both the driver and the truck from ending up in a dangerous situation. It can also be done for various different scenarios and driving conditions but the definition of what is safe is always the same. Safe driving or safe vehicle motion is throughout the paper defined as when the vehicle is not in any of the unsafe modes depicted in Figure 1. The modes includes jack-knifing, rollover, off-tracking and trailer swing. Jack-knifing indicates a scenario where the rear axles of the tractor is losing traction while the trailer maintains traction. Off-tracking is when the steering front axle of the tractor is losing traction, causing the vehicle to deviate from the intended trajectory. Rollover indicates that the vehicle rolls over to its side and trailer swing is when the trailer is losing traction and swings out while turning

¹A. Hansson, ²E. Andersson, ³L. Laine, ⁴M. Sadeghi Kati and ⁵U. Erdinc are with Volvo Group Trucks Technology, SE-405 08 Göteborg, Sweden, axel.e.hansson@gmail.com, erik.andersson.1996@gmail.com, leo.laine@volvo.com, maliheh.sadeghi.kati@volvo.com, umur.erdinc@volvo.com

³L. Laine, ⁵U. Erdinc and ⁶M. Jonasson are with the Department of Mechanics and Maritime Sciences, Div. of Vehicle Engineering and Autonomous Systems, Chalmers University of Technology, Sweden, leo.laine@chalmers.se, umur.erdinc@chalmers.se, mats.jonasson@chalmers.se

while the tractor maintains traction. All four unsafe modes may result in dangerous scenarios which could harm the driver, surrounding people or the vehicle. Creating a control envelope and implementing it into the control system can prevent the vehicle from entering these modes.

The control envelope is obtained through simulation data, described thoroughly in section III. In order to conclude when the simulated vehicle has entered any of the unsafe modes appropriate thresholds are set up for each mode.

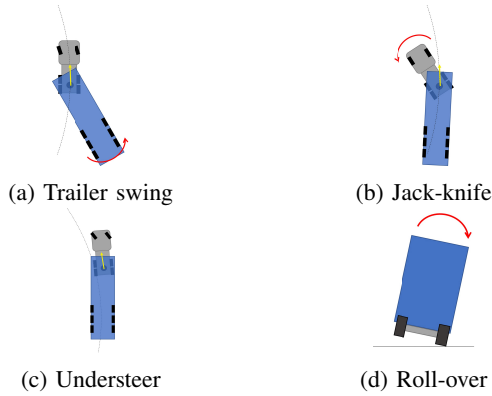


Fig. 1: Unsafe modes of tractor (grey) and semi-trailer (blue) in combination. Red arrows indicates tractor or trailer motion and yellow arrows indicates the coupling point's direction of travel

III. TRAILER SOE

To gather data for the safe operating envelope, a simulation environment must be set up in which a simulated vehicle model can be evaluated in different scenarios and conditions.

A. Data generation

The data for the SOE is generated using simulations on a tractor-semitrailer vehicle unit. The vehicle is run on a test track as depicted in Figure 2. The track is initiated by

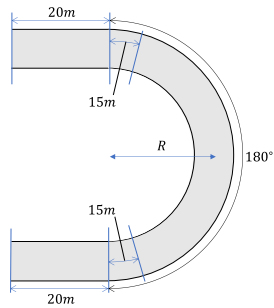


Fig. 2: Test track layout

a straight path of 20 m, followed by a 180° turn with a set radius. The track is ended with another straight of 20 m. Between the straights and the turn, transition parts of 15 m are added to avoid large differences in road curvature which could generate large vehicle motion jerks. The path has no slope or camber and consisted of only one lane. The vehicle is initiated on the road at a set cruise speed which

is controlled by a speed controller which applied enough torque on the two rear axles on the tractor to keep the vehicle at the set speed. Any oscillations in the vehicle states induced by the transition between the initial straight and the curve must first be canceled. This therefore means that the vehicle firstly is run halfway through the track with the speed control acting on the first unit. Then, the speed control is disconnected, letting the vehicle roll without any applied torque. Immediately after the disconnection, a fixed constant torque is applied on each wheel on the two driven front axles on the trailer. This is depicted in Figure 3.

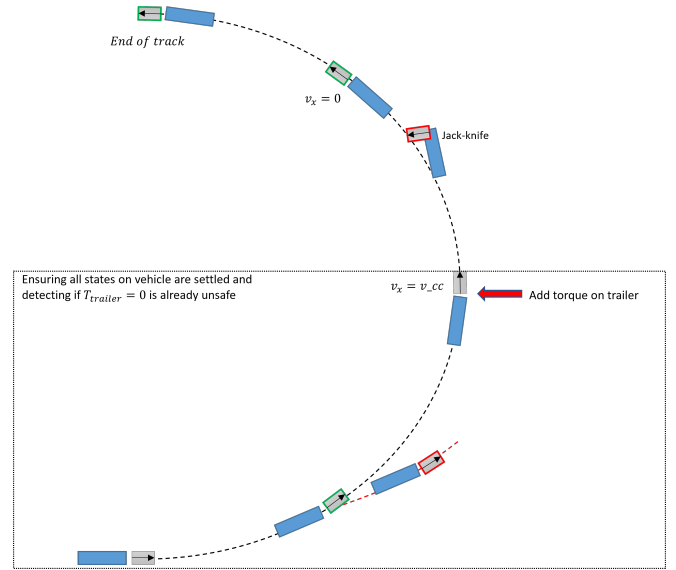


Fig. 3: Simulation test track used to generate data for safe operating envelope

Many simulations with different parameter combinations are run and each simulation produces one data point for the envelope. Depending on the outcome of the simulation, a classification is given to the simulation. These includes for instance the vehicle stopping before reaching the end of the track, any of the unsafe modes described in Section II or that the vehicle reached the end of the track safely.

B. Design of SOE and interpretation of data

By reading and analyzing the states of the vehicle, the parameters of the envelope (certain vehicle states and environmental parameters) can be determined for a certain moment of time. The objective is to compare the current states of the vehicle, hereon denominated as "the state", to a similar already existing simulated point in the control envelope which is classified as safe. When such a point is found, the driver's inputs to the vehicle are limited to the corresponding inputs of the simulated safe point in order to avoid unsafe driving.

The SOE consist of more than 3 parameters which means that visualizing the envelope is impossible without dividing the parameters into several n-dimensional spaces, where $n \in [1 \ 2 \ 3]$. This enables for easy understanding of the inner workings of the envelope. An envelope with for instance 5

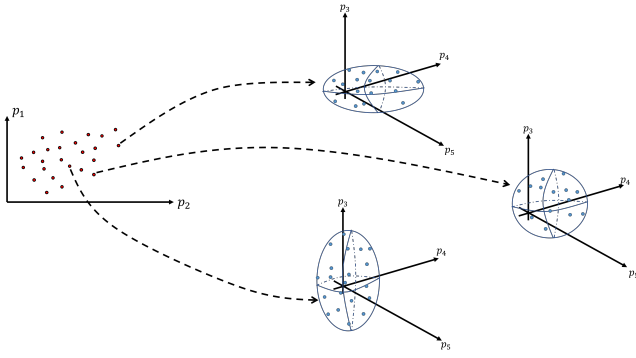


Fig. 4: Envelope parameters spaces concepts

parameters can be visualized by considering it as a set of 3D spaces which are gathered in a 2D space. For each of the data points in the 2D plane there exist a 3D space made up of the remaining parameters. This is depicted in Figure 4 where a point in the plane contains a set of points which are all classified according to how the corresponding simulation went.

In the safe operating envelope assumptions are made about the parameters such that when certain parameter values are increased or decreased, the likelihood of unsafe driving increases or decreases. For instance, higher velocity of the tractor unit is assumed to increase the likelihood of unsafe driving and similar assumptions can be made for all parameters in the envelope. This helps filter away points in the complete envelope which by default can be assumed to be unsafe. To increase the resolution of the envelope and thus increase the performance of the vehicle, certain parameter values are lumped together which are considered close enough to each other. If the envelope does not contain any safe points for the given vehicle state, the system is not limited as there are not enough data points in the envelope.

Algorithm 1 shows the concept of how the SOE can limit the driver inputs in pseudo-code with an arbitrary number of envelope parameters p and driver inputs d .

Algorithm 1 Limiting T using safe operating envelope

Input: $p_1, p_2, p_3, p_4, p_5, E$ $\triangleright$ Input variable E contains envelope
Output: d_1, d_2
 Filter away points which are considered unsafe
 $E \leftarrow$ Remaining points in E with assumptions about p
 $neighbors \leftarrow$ Finds nearest points (neighbors) in E to state
 $n \leftarrow 1$ $\triangleright$ Neighbor index
while $n \leq N_{neighbors}$ **do**
 if $neighbor_n$ is a safe point **then**
 Set and return driver limits in $d_1 d_2$
 else
 $n \leftarrow n + 1$
 end if
end while
if No safe point neighbors were found **then**
 Do not limit driver inputs
else
 Return driver limits $d_1 d_2$
end if

IV. CONTROL SYSTEM

In order for the driver inputs to be limited by the operating envelope and to have the vehicle react and adapt to the limitations, a control system must be established. Here a simplified control system is presented with control allocators together with assisting functions and algorithms to control the vehicle using only driver inputs.

A. Control allocation

Since there are many more actuators on the vehicle than controllable degrees of freedom, the vehicle is recognized to be an over-actuated system. Therefore there exist several different combinations of input signals to the system that can realize the desired behavior of the vehicle. An optimization problem arises which can be solved using a weighted least squares control allocator. This approach enables for finding the optimal control input with regards to limitations of the inputs, prioritization on certain inputs through weight matrices and finally using the dynamics of the studied system through a control efficiency matrix. The WLS problem is formulated as

$$\mathbf{u} = \arg \min_{\mathbf{u} \leq \mathbf{u} \leq \bar{\mathbf{u}}} (\|W_u(\mathbf{u}_{des} - \mathbf{u})\|_2^2 + \gamma \|W_v(B\mathbf{u} - \mathbf{v})\|_2^2) \quad (1)$$

The actuator signals are constrained by their limitations, $\mathbf{u}$ and $\bar{\mathbf{u}}$. W_u and W_v are weighting matrices used to prioritize minimization of certain actuator signals and virtual control signals. $\mathbf{u}_{des}$ are the desired actuator signals and γ is used as a tuning parameter for prioritizing either actuator signals or virtual control signals during the optimization. Figure 5 shows a process diagram of how the control allocators are deployed in the control system.

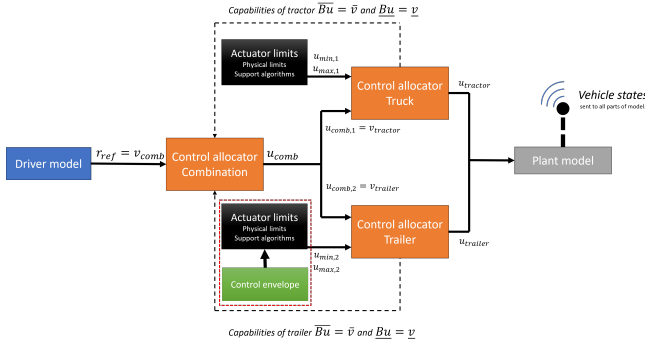


Fig. 5: Control system architecture in which the safe operating envelope is deployed

The combination level control allocator receives inputs in the form of requested global forces on the vehicle from a driver model. It then outputs the distribution of forces on each of the two units to achieve the requested global forces. These are, for a tractor-semitrailer vehicle combination, suitably chosen as

$$\mathbf{v}_{comb} = [F_{x,tot} \quad F_{y,tot} \quad M_{z,1} \quad M_{z,2}] \quad (2)$$

$$\mathbf{u}_{comb} = \begin{bmatrix} F_{x,el,1} & F_{x,sb,1} & F_{y,1} & M_{z,1} & \dots \\ F_{x,el,2} & F_{x,sb,2} & F_{y,2} & M_{z,2} \end{bmatrix} \quad (3)$$

where index 1 refers to the first unit and 2 refers to the second unit. Note that the longitudinal force is also split into two forces, one produced by the electrical motors ('el') (which can be either propelling or braking) and one pure braking force produced by the service brakes ('sb'). Inputs $\mathbf{v}$ to each of the unit control allocators are thus subsets of $\mathbf{u}_{comb}$ and they will output the actuator signals which are then sent to a plant model:

$$\mathbf{v}_{tractor} = [F_{x,el,1} \quad F_{x,sb,1} \quad F_{y,1} \quad M_{z,1}] \quad (4)$$

$$\mathbf{u}_{tractor} = \begin{bmatrix} T_{11L,el} & T_{11R,el} & T_{12L,el} & T_{12r,el} & T_{13L,el} & \dots \\ T_{13R,el} & T_{11L,sb} & T_{11R,sb} & T_{12L,sb} & T_{12R,sb} & \dots \\ T_{13L,sb} & T_{13R,sb} & \delta_{11} \end{bmatrix} \quad (5)$$

$$\mathbf{v}_{trailer} = [F_{x,el,2} \quad F_{x,sb,2} \quad F_{y,2} \quad M_{z,2}] \quad (6)$$

$$\mathbf{u}_{trailer} = \begin{bmatrix} T_{21L,el} & T_{21R,el} & T_{22L,el} & T_{22r,el} & T_{23L,el} & \dots \\ T_{23R,el} & T_{21L,sb} & T_{21R,sb} & T_{22L,sb} & T_{22R,sb} & \dots \\ T_{23L,sb} & T_{23R,sb} \end{bmatrix} \quad (7)$$

Figure 6 shows the forces and definitions of a vehicle unit with trailing and leading units:

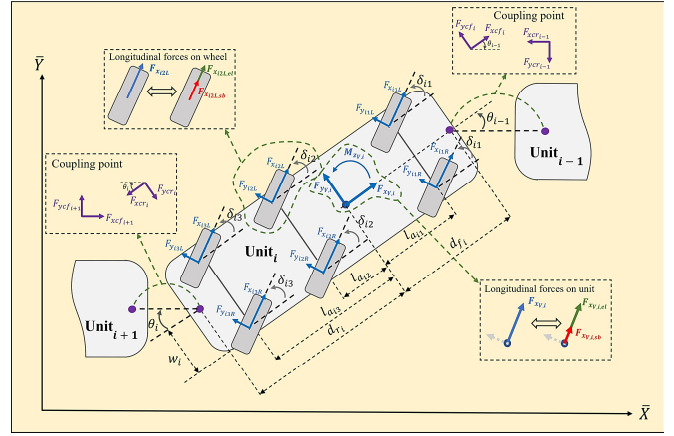


Fig. 6: Definitions of metrics for vehicle unit i with 3 axes and its relations to leading and trailing units

Each vehicle unit provides limits of what upper and lower bounds of $\mathbf{u}$ they are capable of delivering. The capabilities are sent back to the combination level control allocator which re-distributes the global forces on the units. The capabilities are determined in the "Actuator limits"-blocks in Figure 5. The SOE of the trailer set limits on the actuators on the trailer together with other limitations such as physical slip limits of the tyre-road conditions and maximum power output from the actuators on the trailer. Limitations from the SOE are seen as limits that cannot be overridden. The SOE helps create actuator limits on the trailer-unit which can lower or raise the capabilities of the trailer.

B. Control efficiency matrices, B

The control efficiency matrices are responsible for the translation of the driver inputs to the vehicle to the actuator usage through $B\mathbf{u} = \mathbf{v}$. To construct the B-matrices the dynamics of the vehicle must be known and described as a function with respect to the actuator usage. Using Newtonian mechanics, equations of motion of the entire vehicle can be set up as

$$\begin{cases} F_{x1} = m_1 a_{x1} = F_{xV,1,el} + F_{xV,1,sb} - F_{xcr1} - F_{res,1} \\ F_{y1} = m_1 a_{y1} = F_{yV,1} - F_{ygr1} \\ M_{z1} = J_1 \dot{\omega}_{z1} = M_{zV,1} + F_{ygr1} d_{r1} \\ F_{x2} = m_2 a_{x2} = F_{xV,2,el} + F_{xV,2,sb} - F_{xcr1} \cos \theta_1 + F_{ygr1} \sin \theta_1 - F_{res2} \\ F_{y2} = m_2 a_{y2} = F_{yV,2} - F_{xcr1} \sin \theta_1 - F_{ygr1} \cos \theta_1 \\ M_{z2} = J_2 \dot{\omega}_{z2} = M_{zV,2} - F_{xcr1} \sin \theta_1 d_{f2} - F_{ygr1} \cos \theta_1 d_{f2} \end{cases} \quad (8)$$

where index 1 denotes the leading unit and 2 denotes the trailing unit. Complete derivations and equations are found in [11].

It is assumed that there are no non-linearities in the tyre forces and that the actuators are very fast and that no other non-linear dynamics or delays in the driveline of the vehicle exist. These are realistic assumptions for the control of the

vehicle and such the B-matrix for the combination is

$$B = \begin{bmatrix} 1 & 1 & 0 & 0 & \cos \theta_1 & \cos \theta_1 & \sin \theta_1 & 0 \\ 0 & 0 & 1 & 0 & -\sin \theta_1 & -\sin \theta_1 & \cos \theta_1 & 0 \\ 0 & 0 & 0 & 1 & d_{r1} \sin \theta_1 & d_{r1} \sin \theta_1 & -d_{r1} \cos \theta_1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & d_{f2} & 1 \end{bmatrix} \quad (9)$$

Similarly the control efficiency matrices of the two individual units are constructed as showed in Appendix I. The last column of the B-matrix for the tractor is linearized around $\delta_{11} = 0$ as to linearly model the steering forces from the tires of the front axle of the tractor.

C. Energy optimization

Because the vehicle combination system is over-actuated there exist several solutions to the WLS problem that the control allocator solves. Thus the optimal solution can vary depending on the different weighting matrices and tuning parameters: W_u , W_v and γ . One way of pushing the solution in the desired direction is to find a dynamic reference, u_{des} , which is beneficial from an energy perspective. This reference will then be used in the optimization problem when finding the most optimal global force distribution between units, u_{comb} .

The main advantages of using a dynamic reference is that it can be utilized for energy optimization in the control system. Since the vehicle incorporate several separate systems for propulsion and braking, u_{des} can be used as a guidance for the individual units not to counteract each other. Also simultaneous use of the propulsion and braking systems of the same unit can be avoided. This is of particular importance for BEVs since an energy-inefficient system will decrease the total range of the vehicle. For example the service brakes will waste the available kinetic energy as heat, whereas the electric motors instead can be used for recouping energy. A desired prioritization of using the electric motors for braking can also be handled by the dynamic reference.

The reference is generated by an algorithm that uses the actuator capabilities of each unit together with the driver's requested forces, to see if the combination is accelerating or retarding. The forces are then divided equally between the units provided that the corresponding capabilities are not exceeded. If the vehicle is braking the requested forces are primarily allocated to the electric motor braking system and assisted by the service brake system if needed. Ultimately, the solution to the WLS problem in (1) will find the most optimal u_{comb} based on the energy optimized reference u_{des} .

V. RESULTS

Results of using the SOE in conjunction with the control allocation algorithms are presented in this section. The performance of the control system is evaluated by simulating the vehicle combination in two different scenarios, one where only the trailer is active and one where both units are active. The same simulation parameters and 180° turn is used in both scenarios. The road has a constant curvature with a radius of 30 meters and the vehicle's acceleration is handled by a

speed controller acting on the entire combination. The initial longitudinal velocity is $5 \frac{m}{s}$ and the speed controller is set to maintain $25 \frac{m}{s}$. The chosen initial velocity of the vehicle and the curvature of the road ensures that vehicle is able to enter the curve safely, such that off-tracking while entering the curve is avoided.

A. Trailer propulsion only

To verify that the safe operating envelope is able to limit the trailer's propulsion a vehicle combination consisting of a passive tractor (in terms of propulsion and braking) and a propelling trailer is studied. The tractor is only used for steering.

Fig. 7a shows the vehicle's simulated trajectory when the SOE is enabled. The vehicle manages to reach the end of the track safely and both the tractor and trailer travels through the curve as desired. Using the same control system without the SOE results in the trajectory shown in Fig. 7b. Clearly, the trailer has lost traction and started to swing out. Since this is one of the unsafe modes described in section II the simulation is terminated halfway. In Fig. 8 the total actuator usage in both scenarios are displayed. There is a significant difference in signal magnitude between the two cases, where the envelope limits the the signals to roughly half the magnitude of what the system actually requests without it. Limitation of actuator signals also depends of the state of the vehicle. This is what causes the signal to vary over time.

B. Dual unit propulsion

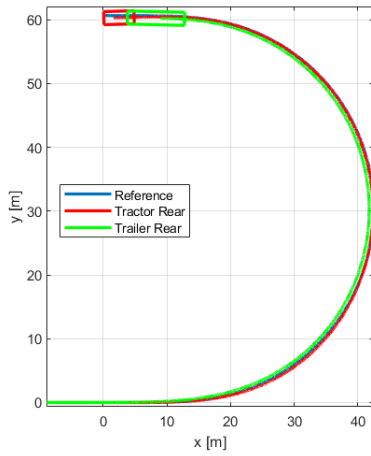
Since the SOE's data points are generated for a vehicle combination where the tractor is passive, one important aspect is to see how the control system performs when both units are active and used for propulsion. In theory the SOE should limit the trailers capability to propel the vehicle, meaning that the tractor's capability will have to be utilized more in order to fulfill the driver's requested global forces.

Fig. 9 shows the trajectories of the simulated vehicle when envelope is both enabled and disabled. The SOE is able to limit the propulsion of the trailer, resulting in successfully reaching the end of the path, as seen in Fig. 9a. However, when the SOE is disabled the vehicle starts off-tracking and ends up in a jack-knifing scenario, as in Fig. 9b.

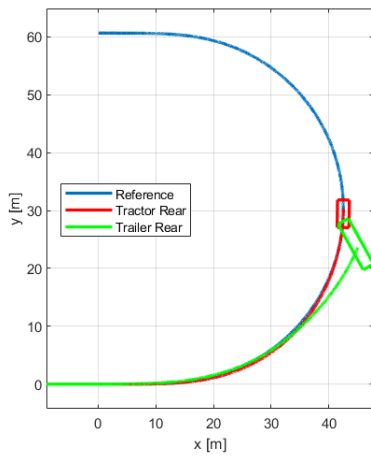
The SOE is thus able to improve the behavior of the vehicle even when two units are used for propulsion. Fig. 10 shows the actuator signals for the case when the envelope is enabled. The sum of the tractor's actuator signals is close to the maximum capability of the tractor where as the SOE is actively limiting the capability and actuator signals of the trailer. Thus the total capability of the vehicle is being limited by the envelope in this scenario.

VI. CONCLUSIONS

A control envelope using experimental data generated off-line implemented in an control architecture based on WLS control allocation proved to avoid a tractor-semitrailer



(a) SOE enabled



(b) SOE disabled

Fig. 7: Complete road path when using only trailer for propulsion, with and without operating envelope. The final pose of the combination is plotted as red (tractor) and green (trailer) rectangles.

combination ending up in unsafe situations. The proposed B-matrices in the allocators which were based on Newtonian mechanics showed a desired distribution of global forces on the vehicle as well as a satisfactory distribution of actuator usage within each unit. The control envelope proved to be a feasible limiting function to total actuator usage on the trailer given that many simulations was run off-line and that the environmental parameters and vehicle states were readily available throughout the use of the envelope. The presented complete control system could with the same methods used in the paper be expanded to include more or fewer, and different connected vehicle units.

The control envelope could include more parameters to further increase the resolution and thus the accuracy of the envelope. A neural network algorithm could also be implemented to expand the envelope during real life driving scenarios of the vehicle, also in order to increase its

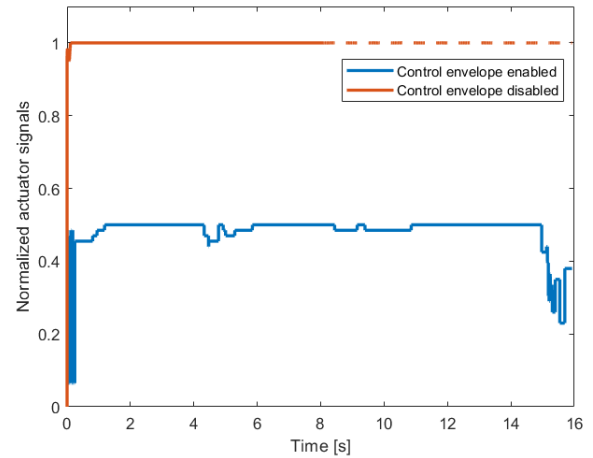
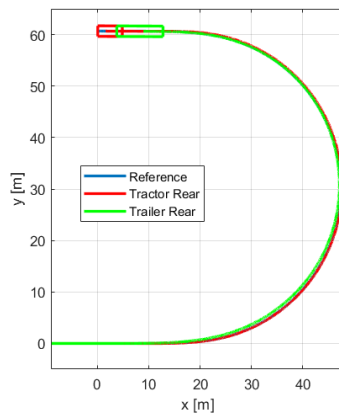


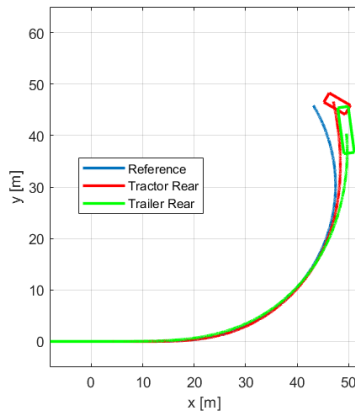
Fig. 8: Actuator usage of trailer with SOE disabled in relation to when SOE is enabled.

resolution.

Since the control system does not have any predictive algorithms the envelope can fail to limit the vehicle in time. Thus model predictive control with a finite horizon could be used as a possible extension of the suggested control system to make the system more reliable and robust.

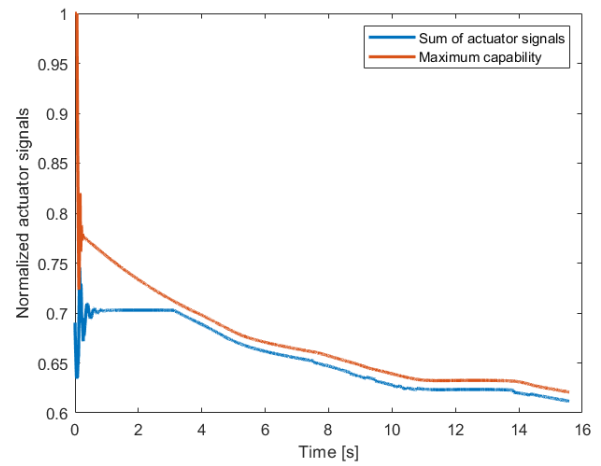


(a) SOE enabled

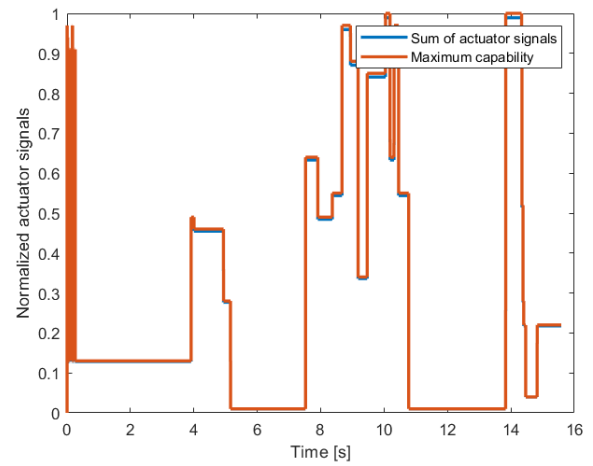


(b) SOE disabled

Fig. 9: Complete road path when using both units for propulsion, with and without SOE. The final pose of the combination is plotted as red (tractor) and green (trailer) rectangles.



(a) Tractor



(b) Trailer

Fig. 10: Actuator signals normalized w.r.t. maximum capability of each unit.

APPENDIX I B-MATRICES

A. Tractor

$$B = \begin{bmatrix} \frac{\cos(\delta_{11})}{R_{11}} & \frac{\cos(\delta_{11})}{R_{11}} & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \cdots & \cdots \\ \frac{\sin(\delta_{11})}{R_{11}} & \frac{\sin(\delta_{11})}{R_{11}} & \cdots & \cdots & \cdots \\ \frac{l_{11} \sin(\delta_{11})}{R_{11}} - \frac{w_{11} \cos(\delta_{11})}{R_{11}} & \frac{w_{11} \cos(\delta_{11})}{R_{11}} + \frac{l_{11} \sin(\delta_{11})}{R_{11}} & \cdots & \cdots & \cdots \\ \frac{1}{R_{12}} & \frac{1}{R_{12}} & \frac{1}{R_{13}} & \frac{1}{R_{13}} & \cdots \\ 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & \cdots \\ -\frac{w_{12}}{R_{12}} & \frac{w_{12}}{R_{12}} & -\frac{w_{13}}{R_{13}} & \frac{w_{13}}{R_{13}} & \cdots \\ 0 & 0 & \cdots & \cdots & \cdots \\ \frac{\cos(\delta_{11})}{R_{11}} & \frac{\cos(\delta_{11})}{R_{11}} & \cdots & \cdots & \cdots \\ \frac{\sin(\delta_{11})}{R_{11}} & \frac{\sin(\delta_{11})}{R_{11}} & \cdots & \cdots & \cdots \\ \frac{l_{11} \sin(\delta_{11})}{R_{11}} - \frac{w_{11} \cos(\delta_{11})}{R_{11}} & \frac{w_{11} \cos(\delta_{11})}{R_{11}} + \frac{l_{11} \sin(\delta_{11})}{R_{11}} & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & 0 & 0 \\ \frac{1}{R_{12}} & \frac{1}{R_{12}} & \frac{1}{R_{13}} & \frac{1}{R_{13}} & 0 \\ 0 & 0 & 0 & 0 & 2Ca_{11} \\ -\frac{w_{12}}{R_{12}} & \frac{w_{12}}{R_{12}} & -\frac{w_{13}}{R_{13}} & \frac{w_{13}}{R_{13}} & 2Ca_{11} l_{11} \end{bmatrix}$$

B. Trailer

$$B = \begin{bmatrix} \frac{1}{R_{21}} & \frac{1}{R_{21}} & \frac{1}{R_{22}} & \frac{1}{R_{22}} & \frac{1}{R_{23}} & \frac{1}{R_{23}} & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ -\frac{w_{21}}{R_{21}} & \frac{w_{21}}{R_{21}} & -\frac{w_{22}}{R_{22}} & \frac{w_{22}}{R_{22}} & -\frac{w_{23}}{R_{23}} & \frac{w_{23}}{R_{23}} & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ \frac{1}{R_{21}} & \frac{1}{R_{21}} & \frac{1}{R_{22}} & \frac{1}{R_{22}} & \frac{1}{R_{23}} & \frac{1}{R_{23}} & \cdots \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots \\ -\frac{w_{21}}{R_{21}} & \frac{w_{21}}{R_{21}} & -\frac{w_{22}}{R_{22}} & \frac{w_{22}}{R_{22}} & -\frac{w_{23}}{R_{23}} & \frac{w_{23}}{R_{23}} & \cdots \end{bmatrix}$$

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